Unsupervised Learning

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Unsupervised Learning

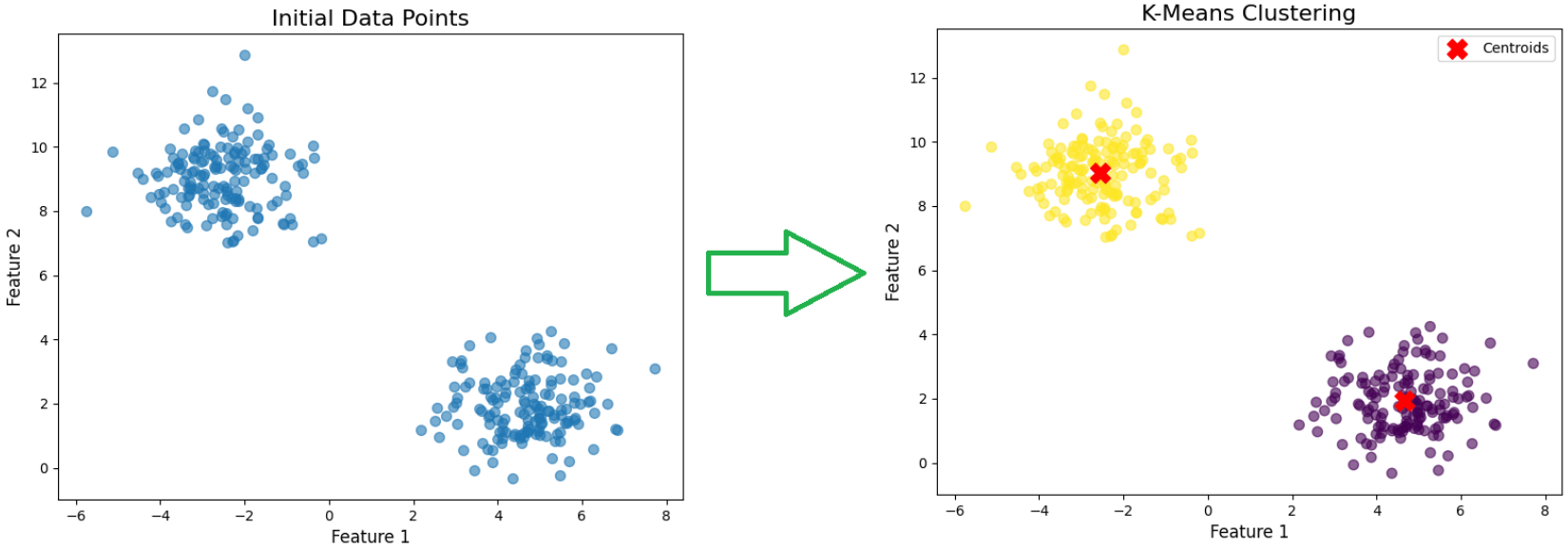
A type of learning where the model works with unlabeled data to discover patterns.

Examples:

- Clustering (e.g., K-Means)
- Dimensionality Reduction (e.g., PCA)

Clustering

- A way of grouping the data points into different clusters, consisting of similar data points
- Objects with possible similarities remain in a group that has less or no similarities with another group

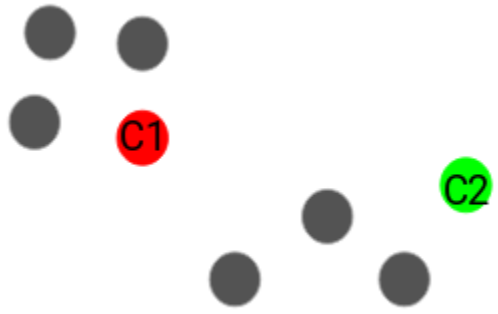


K-means Clustering

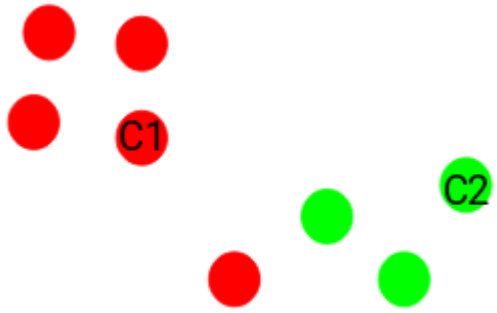
- K-means clustering is a method for grouping ***n*** observations into ***K*** clusters
- Aims to assign each observation to the cluster with the nearest mean or centroid
- We have these 8 points, and we want to apply k-means to create clusters for these points

Intuition

1. Choose the number of clusters k. Let's say it is 2.
2. Select k random points from the data as centroids

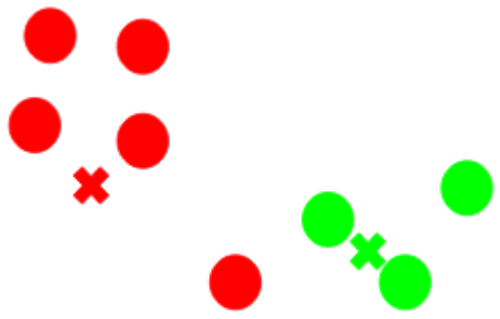


3. Assign all the points to the closest cluster centroid



Intuition (Cont.)

4. Recompute the centroids of newly formed clusters



5. Repeat steps 3 and 4

Stopping Criteria

- Centroids of newly formed clusters do not change
- Points remain in the same cluster
- Maximum number of iterations is reached

Challenges

- Value of k must be predefined
  - A wrong value of k might result in inaccurate result
  - Optimal k can be found using silhouette analysis or elbow method
- The algorithm cannot handle categorical data
  - Distance calculation is inaccurate with categorical features
- High computational cost
- Finds spherical shaped clusters only

Example with Data

- Let’s consider the following data points.
- Point (x, y) A (1,2) B (2,3) C (3,3) D (6,7) E (7,8) F (8,8)

- **Step 1: Choose the number of clusters (k)**
  - We decide to divide the dataset into 2 clusters, k = 2
- **Step 2: Select Initial Centroids Randomly**
  - Centroid 1 (C1) =A (1,2)
  - Centroid 2 (C2) = E (7,8)

Example with Data

- **Step 3: Assign Each Point to the Nearest Centroid**
  - Using Euclidean distance
  - Cluster C1: { A(1,2), B(2,3), C(3,3) }
  - Cluster C2: { D(6,7), E(7,8), F(8,8) }
- **Step 4: Recompute Centroids**
  - The new centroid is the mean of all points in the cluster:
  - C1=($\frac{1+2+3}{3}$,$\frac{2+3+3}{3}$)=(2,2.67)
  - C2=($\frac{6+7+8}{3}$, $\frac{7+8+8}{3}$)=(7,7.67)

Example with Data

- **Step 5: Repeat Steps 3 & 4 Until Convergence**
  - Final Clusters:
  - Cluster 1 (C1): A(1,2), B(2,3), C(3,3)
  - Cluster 2 (C2): D(6,7), E(7,8), F(8,8)

Questions?